

Interactive Visualization in R

& Applications in Business

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Agenda

1. What is Interactive Data Visualization?
2. Why it matters in Business
3. The Boston Housing Dataset as an example
4. Live R Demo with `plotly`
5. Business Applications
6. Class Exercise

What is Interactive Visualization?

An interactive chart allows to perform actions **zooming, hovering data through three dimensions** to reveal patterns invisible in 2D charts

Traditional 2D

- Bar charts
- Line graphs
- Scatter plots
-  Simple, familiar
-  Limited to 2 variables at

3D Visualization

- 3D scatter plots
- Surface plots
- 3D bar charts
-  Shows 3 variables together

When Does Interactive 3D Add Value?

Use Interactive 3D when:

- Three continuous variables must be compared simultaneously.
- Clusters in high-dimensional data need to be revealed.
- An interactive HTML format is available (rotation purposes).

Avoid Interactive 3D when:

- The chart will be printed as a static image.
- Only two variables carry meaningful information.

Figure 1: When to use interactive 3D charts

The Boston Housing Dataset

Source: Harrison and Rubinfeld (1978) — 506 census tracts in Boston

Available in R: MASS package (no download needed!)

	crim	zn	indus	chas	nox	rm	age	dis	rad	tax	ptratio	black	lstat
1	0.00632	18	2.31	0	0.538	6.575	65.2	4.0900	1	296	15.3	396.90	4.98
2	0.02731	0	7.07	0	0.469	6.421	78.9	4.9671	2	242	17.8	396.90	9.14
3	0.02729	0	7.07	0	0.469	7.185	61.1	4.9671	2	242	17.8	392.83	4.03
4	0.03237	0	2.18	0	0.458	6.998	45.8	6.0622	3	222	18.7	394.63	2.94
5	0.06905	0	2.18	0	0.458	7.147	54.2	6.0622	3	222	18.7	396.90	5.33
6	0.02985	0	2.18	0	0.458	6.430	58.7	6.0622	3	222	18.7	394.12	5.21

medv

1	24.0
2	21.6
3	34.7
4	33.4
5	36.2
6	28.7

Variable Description

Why These Three Variables?

Business question:

“How do neighborhood characteristics jointly affect property values?”

- `rm` → quality / size signal
- `lstat` → socioeconomic signal
- `medv` → outcome (price)

A real estate firm, bank, or urban planner would care deeply about this relationship.

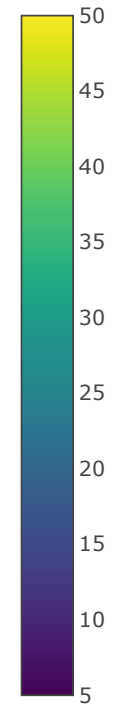
Setting Up in R

```
[1] 506 14
```

rm	lstat	medv
Min. :3.561	Min. : 1.73	Min. : 5.00
1st Qu.:5.886	1st Qu.: 6.95	1st Qu.:17.02
Median :6.208	Median :11.36	Median :21.20
Mean :6.285	Mean :12.65	Mean :22.53
3rd Qu.:6.623	3rd Qu.:16.95	3rd Qu.:25.00
Max. :8.780	Max. :37.97	Max. :50.00

3D Scatter Plot — Code

Boston Housing: Rooms, Status & Home Value



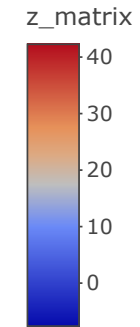
3D Scatter Plot — What We See

Key observations:

- **High *rm* + Low *lstat*** → High home values (yellow cluster at the top, approaching \$50k)
- **Low *rm* + High *lstat*** → Low home values (dark purple cluster at the bottom, around \$10k)
- The **Viridis color scale** reinforces the z-axis — yellow = high value, purple = low value
- The data forms a **diagonal ridge** rising from high *lstat*/low *rm* toward low *lstat*/high *rm*
- A small cluster of **very high value homes** (\$45–50k) appears isolated at low *lstat* and high *rm* suggesting a

3D Surface Plot — Code

Predicted Home Value Surface



Surface Plot — Business Interpretation

The surface shows the **predicted median home value** across all combinations of **rm** and **lstat**

How a business uses this:

Business	Use Case
 Real Estate Firm	Identify undervalued vs premium zones
 Bank / Lender	Assess collateral risk by neighborhood
 Urban Planner	Target investment in low-value areas
 Investment Fund	Screen property acquisition opportunities

Business Applications of 3D Visualization

Beyond real estate:

- **Customer Segmentation** — Age × Income × Spending Score
- **Financial Risk** — Two risk factors × Portfolio loss
- **Supply Chain** — Lead time × Order qty × Cost
- **Marketing** — Campaign spend × Channel × Conversion rate

Important

The key is always the same: use 3D when **three variables together** tell a story that two cannot.

Key Takeaways

- ✓ 3D visualization reveals **multivariate relationships** hidden in 2D
- ✓ `plotly` in R makes it **interactive and easy** to implement
- ✓ The Boston Housing dataset provides a **realistic business context**
- ✓ Always ask: *does the third dimension add genuine insight?*
- ✓ Use **interactive HTML** — never static 3D for print

References

Harrison, David, and Daniel L. Rubinfeld. 1978. "Hedonic Housing Prices and the Demand for Clean Air." *Journal of Environmental Economics and Management* 5 (1): 81–102. [https://doi.org/10.1016/0095-0696\(78\)90006-2](https://doi.org/10.1016/0095-0696(78)90006-2).

Thank You

Questions?

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